UNIVERSITY OF MAURITIUS
SIS 3204Y(5) – NATURAL LANGUAGE PROCESSING

MOCK PAPER 2 – Weeks 1 to 8 (Question Paper with Model Answers)
Total: 50 marks | Suggested duration: 60 minutes

Instructions: Answer ALL questions. Marks are shown beside each question. The week each question comes from is shown in the question header. Model answers are given under each question in bullet points; the mark for a point is shown in brackets, e.g. (1). Cover the answers while practising. Equivalent correct answers should receive credit.

Question 1 | Week 1 (Revision Q8) [7 marks]

Describe the evolution of NLP from rule-based systems to Large Language Models (LLMs).

Answer

- **Rule-based (1950s–1980s):** expert-written grammar rules, dictionaries and pattern matching; struggled with ambiguity. (1)
- **Statistical NLP (1990s):** used probability on large text collections, e.g. N-gram language models. (1)
- **Machine learning (early 2000s):** models learned from labelled data, e.g. Naive Bayes for spam detection. (1)
- **Deep learning (2000s–early 2010s):** neural networks (RNN, LSTM) automated feature extraction and sequence modelling. (1)
- **Word embeddings:** Word2Vec represented word meaning as vectors. (1)
- **Transformers (2017 onwards):** self-attention, processing words in parallel with full context. (1)
- **LLMs (2020 onwards):** very large multi-task models trained on huge text data, e.g. ChatGPT, Claude, Gemini. (1)

Question 2 | Week 2 (Revision Q15) [6 marks]

Explain how subword tokenisation works, using the word “internationalization” as an example. Explain why it handles rare or unknown words better than word-level tokenisation.

Answer

- Subword tokenisation breaks words into **smaller meaningful units** (e.g. WordPiece, BPE) instead of whole words. (1)
- Example: internationalization → inter, ##national, ##ization; a rare word is split into known pieces using the algorithm’s rules. (2)
- Special tokens such as [CLS] and [SEP] may be added for model input. (1)
- Rare/unknown words are built from **known pieces**, whereas word-level tokenisation fails on unknown (OOV) words. (1)
- Other advantages: smaller vocabulary, better generalisation (prefixes, suffixes, roots) and shorter sequences than character-level. (1)

Question 3 | Week 3 (Revision Q7) [6 marks]

Discuss THREE advantages and THREE limitations of one-hot encoding.

Answer

Advantages

- Removes false order and distance between words (no ranking). (1)
- Treats all categories equally (no bias towards bigger or smaller integers). (1)
- Simple, safe and works with linear models, logistic regression and neural networks; each column is easy to interpret. (1)

Limitations

- High dimensionality and sparsity: one column per word, mostly 0s (10,000 words = 10,000 columns). (1)
- Ignores relationships: “cat” and “tiger” look unrelated, no similarity or hierarchy; not scalable for very large vocabularies. (1)
- Higher computational cost (more features = longer training); often unnecessary for tree-based models. (1)

Question 4 | Week 4 (Part C Q7) [4 marks]

Weighted Sum: Given x1 = 2, x2 = 3, w1 = 0.5, w2 = 0.2, b = 0.1. Calculate the weighted sum z and then the ReLU output.

Answer

- Substitution: $z = (0.5 \times 2) + (0.2 \times 3) + 0.1$. (1)
- Weighted products: $0.5 \times 2 = 1.0$ and $0.2 \times 3 = 0.6$. (1)
- $z = 1.0 + 0.6 + 0.1 = \mathbf{1.7}$. (1)
- ReLU(1.7) = max(0, 1.7) = **1.7**. (1)

Question 5 | Week 5 (Sample Q7) [10 marks]

Compare RNN, LSTM and GRU with respect to their memory mechanism, gates, ability to capture long-term dependencies and computational complexity.

Answer

- **RNN:** one hidden state, no gates; simplest architecture with few parameters. (1)
- **RNN limitation:** suffers from vanishing gradients, so it struggles with long-term dependencies. (1)
- **LSTM states:** a hidden state and a separate cell state. (1)
- **LSTM gates:** three gates – forget, input and output. (1)
- **LSTM advantage:** better control of information, so it captures long-term dependencies more effectively. (1)
- **LSTM complexity:** more complex, with more parameters and more computation. (1)
- **GRU:** one hidden state (no separate cell state) and two gates – reset and update. (2)
- **GRU efficiency:** simpler than LSTM, fewer parameters, often trains faster, still captures long-term dependencies well. (1)
- **Overall:** the choice depends on the task, sequence type, computing resources and tested performance. (1)

Question 6 | Week 6 (Set 1 Q7) [8 marks]

For one token, the attention weights are [0.2, 0.5, 0.3]. The corresponding Value vectors are V1 = [1,0], V2 = [0,2] and V3 = [1,1].

Calculate the contextual output vector. You do NOT need to calculate softmax.

Answer

- Output = 0.2·V1 + 0.5·V2 + 0.3·V3 (weighted sum). (2)
- 0.2 × [1,0] = [0.2, 0]. (1)
- 0.5 × [0,2] = [0, 1.0]. (1)
- 0.3 × [1,1] = [0.3, 0.3]. (1)
- Add them: [0.2 + 0 + 0.3, 0 + 1.0 + 0.3] = **[0.5, 1.3]**. (2)
- V2 contributes most because it has the largest attention weight (0.5). (1)

Question 7 | Week 7 (Set 2 Q3) [5 marks]

During pretraining, BERT is given: “The bank approved the [MASK] application.”

a) Name BERT’s principal pretraining objective illustrated by this example. (1 mark)

b) Explain how BERT uses the surrounding words to predict the masked token. (2 marks)

c) A bank wants to categorise incoming emails into “loan enquiry”, “card issue” and “complaint”. Explain how BERT could be used for this task. (2 marks)

Answer

(a)

- **Masked Language Modelling (MLM).** (1)

(b)

- BERT uses bidirectional self-attention, so it can see words **before and after** [MASK]. (1)
- It uses the combined context to predict a suitable word, such as “loan”. (1)

(c)

- Use a pretrained BERT encoder and add a **classification head** (often on [CLS]). (1)
- **Fine-tune** on labelled emails to predict the three categories. (1)

Question 8 | Week 8 (Tutorial Set B Q1) [4 marks]

(a) Explain what makes a foundation model suitable for adaptation to different downstream tasks. (2 marks)

(b) A student claims: “Every foundation model is a large language model.” Is this correct? Justify your answer using one non-text modality. (2 marks)

Answer

(a)

- A foundation model is **pre-trained on broad data** and learns reusable representations. (1)
- It can be adapted or prompted for **many downstream tasks** instead of being trained from scratch for each task. (1)

(b)

- The claim is **incorrect**: foundation models are a wider category; an LLM is specialised in language. (1)
- Example: a foundation model trained for **image** understanding (or audio) is not an LLM. (1)

Marks Summary

Question	Week	Source	Marks
1	1	Revision Q8	7
2	2	Revision Q15	6
3	3	Revision Q7	6
4	4	Part C Q7	4
5	5	Sample Q7	10
6	6	Set 1 Q7	8
7	7	Set 2 Q3	5
8	8	Tutorial Set B Q1	4
		TOTAL	**50**